

GLA UNIVERSITY

Application Build and Release Automation Lab

DIVYANSH NEGI | Roll No. 2415001880 | Semester 5th

QUES 1

```
$ mkdir Divyansh-GLA-LibrarySystem
Directory created successfully.
$ cd Divyansh-GLA-LibrarySystem
Current directory changed to Divyansh-GLA-LibrarySystem.
$ mkdir src docs config backup
Four directories created: src, docs, config, backup.
$ touch library_info.txt
library_info.txt created.
$ printf "Student Name: DIVYANSH NEGI\nRoll Number: 2415001880\nSubject: Application Build and
Release Automation Lab\nProject Name: Online Library Management System\nSemester: 5th\n" >
library_info.txt
Information written to library_info.txt.
$ pwd
/c/Users/DIVYANSH NEGI/Divyansh-GLA-LibrarySystem
$ cd src
Current directory changed to src.
$ cd ..
Returned to Divyansh-GLA-LibrarySystem.
$ ls
backup config docs library_info.txt src
$ cat library_info.txt
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$ tree
.\n    backup\n    config\n    docs\n    library_info.txt\n    src
```

QUES 2

```
$ git init
Initialized empty Git repository in ../Divyansh-GLA-LibrarySystem/.git/
$ git config --global user.name "DIVYANSH NEGI"
Git username configured: DIVYANSH NEGI
$ git config --global user.email "divyanshnegi20@gmail.com"
Git email configured: divyanshnegi20@gmail.com
$ git branch -M main
Primary branch renamed to main.
```

```
$ git status

On branch main\nNo commits yet\nUntracked files: backup/ config/ docs/ library_info.txt src/

$ git add .

All project files staged successfully.

$ git commit -m "Initial commit for library management system"

[main <hash>] Initial commit for library management system\nFiles committed successfully.

$ printf "\nProject Objective: To develop an Online Library Management System.\nProject Description:
The system manages books, members and library records.\n" >> library_info.txt

Project objective and description added to library_info.txt.

$ git status

On branch main\nChanges not staged for commit: library_info.txt

$ git diff

- Project Objective and Project Description lines shown as additions.

$ git add library_info.txt

library_info.txt staged successfully.

$ git commit -m "Add project objective and description"

[main <hash>] Add project objective and description\n1 file changed, 2 insertions(+).

$ git log --oneline

<hash> Add project objective and description\n<hash> Initial commit for library management system
```

QUES 3

```
$ git switch -c book-feature

Switched to a new branch 'book-feature'

$ touch src/book.txt

src/book.txt created.

$ printf "Book Management Module\n\nFeatures:\n1. Add new books\n2. Search books by title or
author\n3. Update and remove book records\n4. Track book availability\n" > src/book.txt

Book Management Module content written successfully.

$ git add src/book.txt

src/book.txt staged successfully.

$ git commit -m "Add book management module"

[book-feature <hash>] Add book management module\n1 file changed, 6 insertions(+).

$ git branch

* book-feature\n main

$ git branch --show-current

book-feature

$ git log book-feature --oneline

<hash> Add book management module\n<hash> Add project objective and description\n<hash> Initial
commit for library management system

$ git switch main

Switched to branch 'main'
```

QUES 4

```
$ git switch main

Already on 'main'

$ git merge book-feature

Updating <hash>..<hash>\nFast-forward\n src/book.txt | 6 +++++\n1 file changed, 6 insertions(+).

$ ls src

book.txt

$ cat src/book.txt

Book Management Module\n\nFeatures:\n1. Add new books\n2. Search books by title or author\n3. Update
and remove book records\n4. Track book availability

$ git switch -c library-update

Switched to a new branch 'library-update'

$ sed -i 's/Project Name: Online Library Management System/Project Name: Library Management System -
Branch Update/' library_info.txt

library_info.txt updated on library-update.

$ git add library_info.txt

library_info.txt staged successfully.

$ git commit -m "Update project name in library branch"
```

```

[library-update <hash>] Update project name in library branch\n1 file changed, 1 insertion(+), 1
deletion(-).

$ git switch main

Switched to branch 'main'

$ sed -i 's/Project Name: Online Library Management System/Project Name: Online Library Management
System - Main Update/' library_info.txt

library_info.txt updated on main.

$ git add library_info.txt

library_info.txt staged successfully.

$ git commit -m "Update project name in main branch"

[main <hash>] Update project name in main branch\n1 file changed, 1 insertion(+), 1 deletion(-).

$ git merge library-update

Auto-merging library_info.txt\nCONFLICT (content): Merge conflict in library_info.txt\nAutomatic
merge failed; fix conflicts and then commit the result.

$ git status

On branch main\nYou have unmerged paths.\n both modified: library_info.txt

$ cat library_info.txt

<<<<<< HEAD\nProject Name: Online Library Management System - Main Update\n=====\nProject Name:
Library Management System - Branch Update\n>>>>>> library-update

$ sed -i '/^<<<<<< /d; /^=====/d; /^>>>>>> /d' library_info.txt

Conflict markers removed from library_info.txt.

$ sed -i 's/Project Name:.* /Project Name: Online Library Management System - Final Resolved Version/'
library_info.txt

Conflict resolved with final project name.

$ git add library_info.txt

Resolved file staged successfully.

$ git commit -m "Resolve library update merge conflict"

[main <hash>] Resolve library update merge conflict\nMerge conflict resolution committed.

$ cat library_info.txt

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$ git status

On branch main\nnothing to commit, working tree clean

$ git log --oneline --graph --all

* <hash> Resolve library update merge conflict\n|\n| * <hash> Update project name in library
branch\n| * <hash> Update project name in main branch\n|/\n* <hash> Add book management module\n*
<hash> Add project objective and description\n* <hash> Initial commit for library management system

```

QUES 5

```
$ tree
.\n■■■■ backup\n■■■■ config\n■■■■ docs\n■■■■ library_info.txt\n■■■■ src\n■■■■ book.txt

$ git status

On branch main\nnothing to commit, working tree clean

$ git branch

book-feature\n* main\n library-update

$ git branch --show-current

main

$ git log --oneline --graph --all

* <hash> Resolve library update merge conflict\n|\\\n| * <hash> Update project name in library
branch\n* | <hash> Update project name in main branch\n|/\n* <hash> Add book management module\n*
<hash> Add project objective and description\n* <hash> Initial commit for library management system

$ cat src/book.txt

Book Management Module\n\nFeatures:\n1. Add new books\n2. Search books by title or author\n3. Update
and remove book records\n4. Track book availability

$ cat library_info.txt

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```